

MATH 301: Homework 5

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April 11, 2026

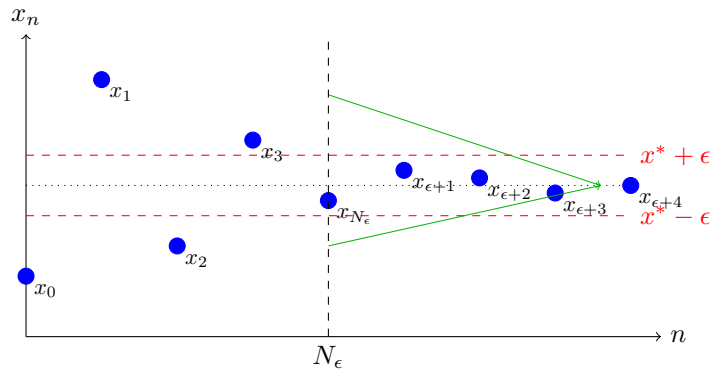
Cauchy Definition A sequence $X = (x_n)$ is called a *Cauchy sequence* if the terms of the sequence get arbitrarily close to each other as the sequence progresses. More formally,

For every $\epsilon > 0$, there exists an integer $N_\epsilon > 0$, such that

$$\forall m, n \geq N_\epsilon, \quad |x_n - x_m| < \epsilon.$$

This means that after some index N_ϵ , the distance between any two terms x_n and x_m (with n, m large enough) is less than any chosen positive number ϵ . Intuitively, the sequence becomes "clustered" together and the terms never drift far apart, no matter how far out in the sequence you go.

Example of a Cauchy Sequence



Section 3.5: 2b Show directly from the definition that the following is a Cauchy sequence:

$$x_n := \left(1 + \frac{1}{2!} + \cdots + \frac{1}{n!} \right)$$

Intuitive English: x_n is Cauchy if there is some index N_ϵ such that given any two indices $m, k \geq N_\epsilon$, the difference between the two terms is "sufficiently small": $|x_m - x_k| < \epsilon$.

Define terms:

$$x_m := \sum_{i=m}^{\infty} \frac{1}{i!} \quad x_k := \sum_{i=k}^{\infty} \frac{1}{i!}$$

Goal (Show the Following):

$$\forall \epsilon > 0, \quad \left| \sum_{i=m}^{\infty} \frac{1}{m!} - \sum_{i=k}^{\infty} \frac{1}{k!} \right| < \epsilon$$

$$\equiv \left| 1 + \frac{1}{2!} + \frac{1}{m!} - 1 - \frac{1}{2!} - \frac{1}{k!} \right| < \epsilon$$

Define terms:

$$x_m := \sum_{i=m|N_\epsilon < m}^{\infty} \frac{1}{m!}$$

$$x_k := \sum_{i=k|N_\epsilon < k}^{\infty} \frac{1}{n!}$$

Goal (Show the Following):

$$\forall \epsilon > 0, \left| \sum_{i=m|N_\epsilon < m}^{\infty} \frac{1}{m!} - \sum_{i=k|N_\epsilon < k}^{\infty} \frac{1}{n!} \right| < \epsilon$$

$$\equiv \left| 1 + \frac{1}{2!} + \cdots + \frac{1}{m!} - \left(1 + \frac{1}{2!} + \cdots + \frac{1}{n!} \right) \right| < \epsilon$$

Suppose $n < m$:

$$\left| \frac{1}{(n+1)!} + \cdots + \frac{1}{m!} \right| \leq \left| \frac{1}{2^{n+1}} + \frac{1}{2^{n+2}} + \cdots + \frac{1}{2^m} \right|$$

$$\equiv \sum_{k=n+1}^{\infty} \left(\frac{1}{2} \right)^k$$

Evaluating the geometric series:

$$\sum_{k=n+1}^{\infty} \left(\frac{1}{2} \right)^k = \frac{\frac{1}{2^{n+1}}}{1 - \frac{1}{2}} = \frac{1}{2^n}$$

$$-\frac{1}{n} \leq \frac{1}{2^{n-1}} \leq \frac{1}{n} \implies \lim_{n \rightarrow \infty} \frac{1}{2^{n-1}} = 0$$

$$-\frac{1}{2^{n-1}} \leq |x_m - x_n| \leq \frac{1}{2^{n-1}} \implies \lim_{n \rightarrow \infty} |x_m - x_n| = 0$$

$$\sum_{k=n+1}^{\infty} \left(\frac{1}{2} \right)^k = \frac{\frac{1}{2^n}}{1 - \frac{1}{2}} = \frac{1}{2^{n-1}}$$

$$-\frac{1}{n} \leq \frac{1}{2^{n-1}} \leq \frac{1}{n} \implies \lim_{n \rightarrow \infty} \frac{1}{2^{n-1}} = 0$$

$$-\frac{1}{2^{n-1}} \leq |x_m - x_n| \leq \frac{1}{2^{n-1}} \implies \lim_{n \rightarrow \infty} |x_m - x_n| = 0$$

$$\lim |x_m - x_n| = 0 \implies \forall \epsilon > 0, \exists N_\epsilon > 0, \forall m, n \geq N_\epsilon, |x_n - x_m| < \epsilon. \sum_{k=n+1}^{\infty} \left(\frac{1}{2} \right)^k = \frac{1}{2^{m-n}}$$

$$0 < k := m - n \quad \frac{1}{2^k}$$

$$-\frac{1}{k} < \frac{1}{2^k} < \frac{1}{k} \implies \lim \frac{1}{2^k} = 0$$

■

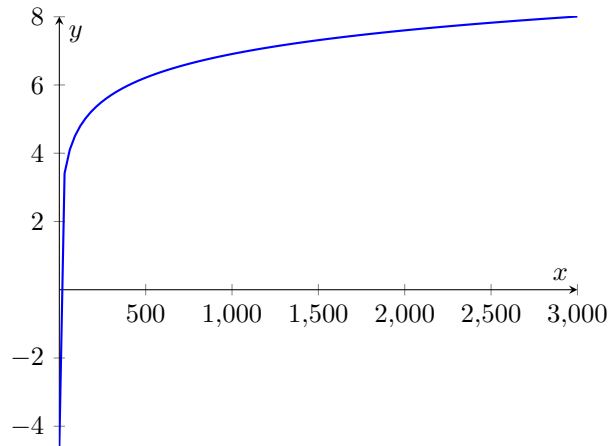
Section 3.5: 3c Show directly from the definition that the following is NOT a Cauchy sequence:

$$\ln(n)$$

Negation of Cauchy

$$\exists \epsilon > 0 \text{ s.t. } \forall n > 0, \exists m, k \geq n \text{ s.t. } |x_n - x_m| \geq \epsilon.$$

Graph of $\ln(x)$



Let the subsequences be defined as:

$$x_m := \ln(2n) \quad \text{and} \quad x_n := \ln(n)$$

Consider the difference:

$$\ln(2n) - \ln(n) = \ln\left(\frac{2n}{n}\right) = \ln(2) \approx 0.6931$$

Since ϵ can be chosen arbitrarily, let $\epsilon \in (0, 1]$. This implies:

$$\exists \epsilon \text{ s.t. } 0.6931 > \epsilon$$

$\therefore x_n$ is not Cauchy.

Section 3.5: 4 Show directly from the definition that if (x_n) and (y_n) are Cauchy sequences, then $(x_n + y_n)$ and $(x_n y_n)$ are Cauchy sequences.

Let the sequence operations be defined as:

$$x_n + y_n := \{x_n + y_n \mid \forall n\} \quad \text{and} \quad x_n \cdot y_n := \{x_n \cdot y_n \mid \forall n\}$$

By the definition of Cauchy sequences:

$$\begin{aligned} |x_m - x_n| < \epsilon &\implies \forall \epsilon > 0, \exists n_\epsilon \text{ s.t. } \forall m, n > n_\epsilon, |x_m - x_n| < \epsilon \\ |y_m - y_n| < \epsilon &\implies \forall \epsilon > 0, \exists n_\epsilon \text{ s.t. } \forall m, n > n_\epsilon, |y_m - y_n| < \epsilon \end{aligned}$$

For addition:

$$|x_m + y_m - (x_n + y_n)| = |(x_m - x_n) + (y_m - y_n)| \leq |x_m - x_n| + |y_m - y_n| < 2\epsilon$$

For multiplication, we establish bounds:

$$\begin{aligned} |x_m - x_n| &\leq |x_m| + |-x_n| \implies |y_m||x_m - x_n| \leq (|x_m| + |-x_n|)|y_m| \\ |y_m - y_n| &\leq |y_m| + |-y_n| \implies |x_n||y_m - y_n| \leq (|y_m| + |-y_n|)|x_n| \end{aligned}$$

Now applying the triangle inequality trick for multiplication:

$$\begin{aligned} |x_my_m - x_ny_n| &= |x_my_m - \mathbf{x_m y_n} + \mathbf{x_m y_n} - x_ny_n| \\ &= |x_m(y_m - y_n) + y_n(x_m - x_n)| \\ &\leq |x_m||y_m - y_n| + |y_n||x_m - x_n| \end{aligned}$$

Because Cauchy sequences are bounded, we know:

$$\begin{aligned} |x_m - x_n| < \epsilon &\implies \exists M_x > 0 \text{ s.t. } \forall m, |x_m| < M_x \\ |y_m - y_n| < \epsilon &\implies \exists M_y > 0 \text{ s.t. } \forall m, |y_m| < M_y \end{aligned}$$

Multiplying by these bounds:

$$\begin{aligned} M_x \cdot |y_m - y_n| &< \epsilon \cdot M_x \\ M_y \cdot |x_m - x_n| &< \epsilon \cdot M_y \end{aligned}$$

Thus:

$$\begin{aligned} \implies |x_my_m - x_ny_n| &< M_x|y_m - y_n| + M_y|x_m - x_n| \\ \text{Letting } \epsilon' = \frac{\epsilon}{2M_x} \text{ and } \epsilon'' = \frac{\epsilon}{2M_y} \text{ yields:} \\ M_x \left(\frac{\epsilon}{2M_x} \right) + M_y \left(\frac{\epsilon}{2M_y} \right) &= \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon \end{aligned}$$

$$x_n := \ln(n)$$

If x_n is a Cauchy sequence. Then $x_n \rightarrow L$ as $n \rightarrow \infty$.

Section 3.5: 7 Let (x_n) be a Cauchy sequence such that x_n is an integer for every $n \in \mathbb{N}$. Show that (x_n) is ultimately constant.

Let (x_n) be a Cauchy sequence where $x_n \in \mathbb{Z}$ for all $n \in \mathbb{N}$.

By the definition of a Cauchy sequence, choosing $\epsilon = 1$, there exists an $N \in \mathbb{N}$ such that for all $m, n \geq N$:

$$|x_m - x_n| < 1$$

Since $x_m, x_n \in \mathbb{Z}$, their absolute difference $|x_m - x_n|$ must be a non-negative integer.

The only non-negative integer strictly less than 1 is 0. Therefore, for all $m, n \geq N$:

$$|x_m - x_n| = 0 \implies x_m = x_n$$

Thus, for all $n \geq N$, $x_n = x_N$. The sequence is ultimately constant.

Section 3.6: 3 Show directly from the definition that the following are not Cauchy sequences.

$$(a) ((-1)^n) \quad (b) \left(n + \frac{(-1)^n}{n}\right) \quad (c) (\ln(n))$$

Observe:

$$\forall n \in \mathbb{N}, |(-1)^n - (-1)^{n+1}| = 2 \geq \epsilon,$$

For some fixed n let $m := 2n$ s.t.

$$x_m := \left(2n + \frac{(-1)^{2n}}{2n}\right) \quad x_n := \left(n + \frac{(-1)^n}{n}\right)$$

Therefore:

$$\begin{aligned} |x_m - x_n| &\equiv \left| \left(2n + \frac{(-1)^{2n}}{2n}\right) - \left(n + \frac{(-1)^n}{n}\right) \right| \equiv \left| \left(2n + \frac{1}{2n}\right) - \left(n + \frac{(-1)}{n}\right) \right| \\ &\equiv \left| \left(2n + \frac{1}{2n}\right) - \left(n + \frac{(-1)}{n}\right) \right| \equiv \left| n + \frac{1}{2n} + \frac{1}{n} \right| \\ n + \frac{1}{2n} + \frac{1}{n} &< n \text{ and } \forall n \in \mathbb{N}, \epsilon \leq |n - L| \implies \epsilon \leq 1 \leq \left| n + \frac{1}{2n} + \frac{1}{n} \right| \\ \therefore \epsilon &\leq |x_m - x_n| \implies x_n \text{ is not Cauchy} \end{aligned}$$

Let $x_n := N$, where $N \in \mathbb{N}$ for an arbitrary integer. Consider $\forall m, n \in \mathbb{N}$:

$$|x_m - x_n| \equiv |N - N| \equiv |0| = 0 < \epsilon$$

Since ϵ can be chosen as $1/2$, the difference between any two integer terms must be strictly less than $1/2$, which is only possible if they are exactly equal (i.e., $|0|$). Thus, the sequence is ultimately constant.

Section 3.6: 7 Let (x_n) and (y_n) be sequences of positive numbers such that $\lim(x_n/y_n) = 0$.

- (a) Show that if $\lim(x_n) = +\infty$, then $\lim(y_n) = +\infty$.
- (b) Show that if (y_n) is bounded, then $\lim(x_n) = 0$.

Infinite Limit: Given $\lim\left(\frac{x_n}{y_n}\right) = 0$ and $x_n, y_n > 0$. Choose $\epsilon = 1$. $\exists N_1$ such that $n \geq N_1 \implies 0 < \frac{x_n}{y_n} < 1 \implies x_n < y_n$.

$\lim x_n = +\infty \implies \exists N_2 : n \geq N_2 \implies x_n > M$. Let $N = \max(N_1, N_2)$. Then $n \geq N \implies y_n > x_n > M$. So, $\lim y_n = +\infty$.

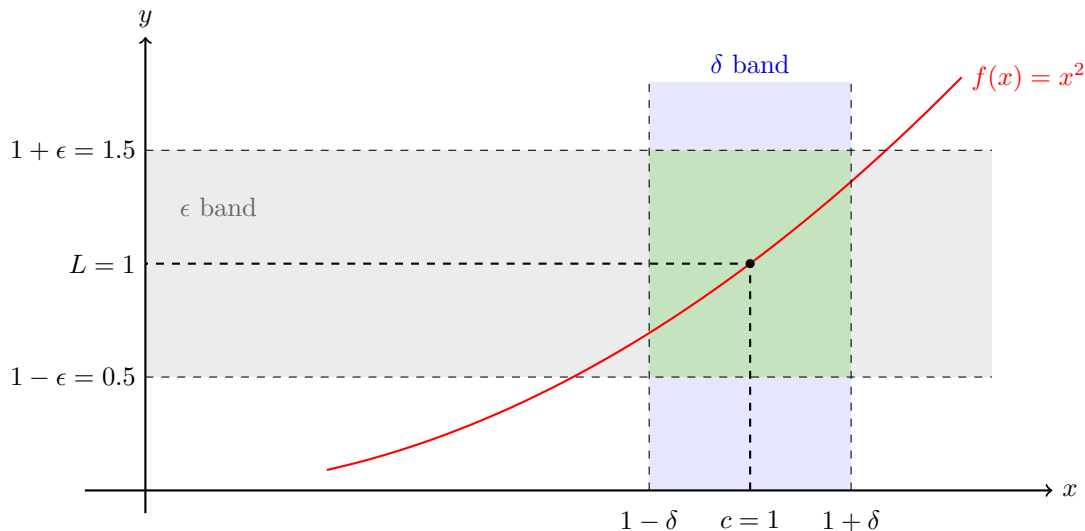
Bounded y_n : Suppose $0 < y_n \leq M \forall n$. For $\epsilon > 0$, $\exists N$ such that $n \geq N \implies 0 < \frac{x_n}{y_n} < \frac{\epsilon}{M}$, so $x_n < \frac{\epsilon}{M} y_n \leq \epsilon$. Thus, $n \geq N \implies 0 < x_n < \epsilon \implies \lim x_n = 0$.

- (a) There exists N_1 such that if $n > N_1$, then $0 < x_n < y_n$. If $\lim(x_n) = \infty$ then $\lim(y_n) = \infty$.
- (b) Suppose that $|y_n| \leq M$ for some $M > 0$. Given $\varepsilon > 0$ there exists N_ε such that if $n \geq N_\varepsilon$ then $0 < \frac{x_n}{y_n} \leq \frac{\varepsilon}{M}$. Therefore $|x_n| \leq (\varepsilon/M)y_n \leq \varepsilon$ for $n \geq N_\varepsilon$.

Section 4.1: 1(a) Determine a condition on $|x - 1|$ that ensures:

$$|x^2 - 1| < \frac{1}{2}$$

Visualization



So any values of $f(x)$ where $x \in (1 - \delta, 1 + \delta)$ should be in the green region.

In other words: $|x - 1| \implies c = 1$, where c represents the limit/cluster point where x approaches. These x 's are defined as: $0 < |x - c| < \delta$ or in other words, for every input to the function, x , where $x \in (c - \delta, c + \delta)$ and x is not necessarily in the domain of the function.

We are interested in the behavior of $f(x) := x^2 - 1$ as $x \rightarrow c$. Specifically, given $\epsilon = 1/2$. Find a corresponding δ s.t. $|f(x) - L| < \epsilon$ where $L = 1$.

Putting it together, given a cluster point $c = 1$ and the function $f(x) := x^2 - 1$ find a corresponding δ s.t. $\forall x \in (c - \delta, c + \delta)$, the values of the function, $f(x) \in (L - \epsilon, L + \epsilon)$ where the limit is defined to be $L := 1$

Recall, are constrained to values of d we can pick from $c = 1$. First define the values of x we are concerned about.

Observe:

$$|x - 1| \leq 1 \equiv |x - 1 + 2| \leq 1 + 2 \equiv |x + 1| \leq 3$$

Therefore any values of x which satisfy the inequality above contain the values which we must justify the inequality $|f(x) - 1| < \frac{1}{2}$.

Observe:

$$x^2 - 1 = (x - 1)(x + 1)$$

difference of squares

Re-establish Inequality

$$|x^2 - 1| = |x - 1| |x + 1| \implies |x - 1| |x + 1| \leq 3 |x - 1| \text{ Multiply both sides of } |x + 1| \leq 3 \text{ by } |x - 1|$$

As $f(x) := x^2$ is greater than or equal to $f'(x) = x$ showing the larger function $f'(x) \rightarrow 1$ shows the smaller function $f(x) \rightarrow 1$ as $x \rightarrow c$.

$$3|x-1| \leq \frac{1}{2} \implies |x-1| < \frac{1}{6}$$

Therefore, given $|x-1| < \frac{1}{6}$ is satisfied we can guarantee the less-strict bound is holds (i.e.)

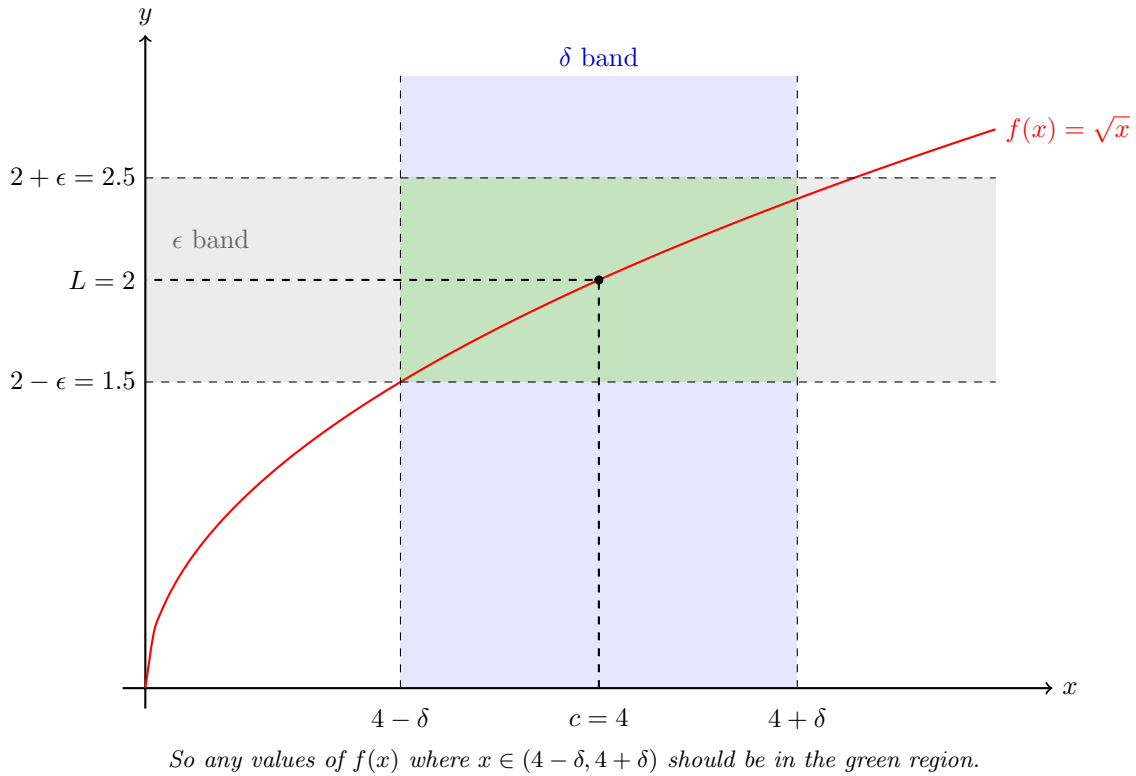
$$\therefore |x-1| < \frac{1}{6} \implies |x^2-1| < \frac{1}{2}$$

Thus our δ should be

Section 4.1: 2(a) Determine a condition on $|x-4|$ that ensures:

$$|\sqrt{x}-2| < \frac{1}{2}$$

Visualization



Find x_ϵ

$$|f(x_\delta) - L| < \epsilon \implies |\sqrt{x_\delta} - 2| < \frac{1}{2} \implies -\frac{1}{2} < \sqrt{x_\delta} - 2 < \frac{1}{2} \implies \frac{3}{2} < \sqrt{x_\delta} < \frac{5}{2} \implies \frac{9}{4} < x_\delta < \frac{25}{4}$$

Find δ_ϵ

$$|x_\delta - c| < \delta \implies |x_\delta - 4| < \delta \implies -\delta < x_\delta - 4 < \delta$$

Therefore:

$$\frac{9}{4} < x_\delta < \frac{25}{4} \equiv -\delta + 4 < x_\delta < \delta + 4$$

$$\begin{aligned} \implies \frac{9}{4} - 4 &< x_{\delta} - 4 < \frac{25}{4} - 4 \\ \implies -\frac{7}{4} &< x_{\delta} - 4 < \frac{9}{4} \end{aligned}$$

Thus we can constrain the equation $f(x) \rightarrow 1/2$ as $x \rightarrow 4$ for any $x \in (4 - \frac{7}{4}, 4 + \frac{9}{4})$

Section 4.1: 4 If $\lim_{y \rightarrow c} f(y) = L$. Then $\forall \epsilon > 0, \exists \delta > 0$ s.t. $0 < |y - c| < \delta \implies |f(y) - L| < \epsilon$.
Let $x := y - c$ s.t. $y = x + c$. Conclude $\lim_{x \rightarrow 0} f(x + c) = L$

$$\begin{aligned} \lim_{y \rightarrow c} f(y) = L &\implies \forall \epsilon > 0, \exists \delta > 0 \text{ s.t.} \\ 0 < |y - c| < \delta &\implies |f(y) - L| < \epsilon \end{aligned}$$

Consider:

$$\begin{aligned} 0 < |y - c_y| < \delta &\implies \\ \exists (y_n) \text{ s.t. } \forall n > n_{\epsilon}, &|y_n - c_y| < \delta \\ y_n &:= x_n + c_y \end{aligned}$$

$$|x_n + c - c| < \delta \implies c_x := 0$$

Section 4.1: 6 Let I be an interval in $\mathbb{R}$. Let $f : I \rightarrow \mathbb{R}$ Let $c \in I$. Suppose $\exists K, L$ s.t. $\forall x \in I, |f(x) - L| \leq K|x - c|$ Show $\lim_{x \rightarrow c} f(x) = L$

Recall:

$$\begin{aligned} \lim_{x \rightarrow c} f(x) = L &\implies \\ \forall \epsilon > 0, \exists n_{\epsilon} > 0 &\text{ s.t.} \\ \forall m, n \geq n_{\epsilon}, &|x_n - x_m| < \epsilon \end{aligned}$$

We must show:

$$\forall \epsilon = K|x - c|, \exists \delta_{\epsilon} \text{ s.t. } \delta_{\epsilon} := 0 < |x - c| < \delta_{\epsilon} \wedge c \notin I$$

Observe

$$\epsilon = K|x - c| \equiv \epsilon = K\delta \implies \delta = \frac{K}{\epsilon} \equiv \delta = Kn$$

$$|x - c| < \delta \equiv -\delta < x - c < \delta \equiv -Kn < x - c < Kn$$

By archimedian property $\exists n \in \mathbb{N}$ s.t. the inequality holds.

How to Pick Which to Use?

Sequential criterion for limits The sequential criterion is useful when it's easier to reason about the behavior of all sequences converging to a point, rather than working directly with ϵ and δ . Use the sequential criterion when:

- The function is complicated, or piecewise-defined, making ϵ - δ algebraic manipulations messy.
- You wish to prove a limit does *not* exist (by producing sequences approaching the point with

different limit values).

The ε - δ definition The ε - δ definition is the standard and most direct way to prove limits exist. Use the ε - δ approach when:

- The function is well-behaved and algebra allows you to solve for δ in terms of ε .
- The problem asks explicitly for an ε - δ proof, or for instructive calculation of how δ depends on ε .

The Sequential Criterion for Limits

Theorem 1 (Equivalence of Epsilon-Delta and Sequential Limits). *Let $A \subseteq \mathbb{R}$, let $f : A \rightarrow \mathbb{R}$ be a function, and let $c \in \mathbb{R}$ be a cluster point (or limit point) of A . Then the following two statements are logically equivalent:*

1. **The Functional Limit:** $\lim_{x \rightarrow c} f(x) = L$
2. **The Sequential Limit:** For every sequence (x_n) in A such that $x_n \neq c$ for all $n \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} x_n = c$, the sequence of function values $(f(x_n))$ converges to L :

$$\lim_{n \rightarrow \infty} f(x_n) = L$$

Section 4.1: 9(a,c) Use either the $\varepsilon - \delta$ definition or the Sequential Criterion for limits to establish the following limits:

$$\lim_{x \rightarrow 2} \frac{1}{1-x} = -1 \quad \lim_{x \rightarrow 0} \frac{x^2}{|x|} = 0$$

Let $\varepsilon > 0$ be given. We must find a $\delta > 0$ such that if $0 < |x - 2| < \delta$, then $\left| \frac{1}{1-x} - (-1) \right| < \varepsilon$.

First, we simplify the expression:

$$\left| \frac{1}{1-x} - (-1) \right| = \left| \frac{1}{1-x} + 1 \right| = \left| \frac{1+1-x}{1-x} \right| = \left| \frac{2-x}{1-x} \right| = \frac{|x-2|}{|x-1|}$$

Next, we establish an initial bound. Assume $\delta \leq \frac{1}{2}$. Then:

$$\begin{aligned} |x-2| &< \frac{1}{2} \\ -\frac{1}{2} &< x-2 < \frac{1}{2} \\ \frac{3}{2} &< x < \frac{5}{2} \\ \frac{1}{2} &< x-1 < \frac{3}{2} \end{aligned}$$

Taking the reciprocal, we get:

$$\frac{1}{|x-1|} < 2$$

Substituting this back into our simplified expression, we have:

$$\frac{|x-2|}{|x-1|} < 2|x-2|$$

We want this entire expression to be less than ϵ . Therefore, we require:

$$2|x - 2| < \epsilon \implies |x - 2| < \frac{\epsilon}{2}$$

Choose $\delta = \min\left(\frac{1}{2}, \frac{\epsilon}{2}\right)$.

Then, for all x satisfying $0 < |x - 2| < \delta$, it follows that $|x - 2| < \frac{1}{2}$ (which guarantees $\frac{1}{|x-1|} < 2$) and $|x - 2| < \frac{\epsilon}{2}$.

Thus:

$$\left| \frac{1}{1-x} - (-1) \right| = \frac{|x-2|}{|x-1|} < 2|x-2| < 2\left(\frac{\epsilon}{2}\right) = \epsilon$$

■

Let $\epsilon > 0$ be given. We must find a $\delta > 0$ such that if $0 < |x - 0| < \delta$, then $\left| \frac{x^2}{|x|} - 0 \right| < \epsilon$.

Observe as $0 < |x - 0| < \delta$, we know $x \neq 0$.

$$\left| \frac{x^2}{|x|} - 0 \right| < \epsilon \implies \frac{x^2}{|x|} < \epsilon \implies \frac{|x^2|}{|x|} < \epsilon \wedge x \neq 0 \implies |x| < \epsilon$$

■

Section 4.1: 12(b,d) Show that the following limits do *not* exist

$$(b) \quad \lim_{x \rightarrow 0^+} \frac{1}{\sqrt{x}} \qquad (d) \quad \lim_{x \rightarrow 0} \sin\left(\frac{1}{x^2}\right)$$

Recall: By **Sequential Criterion** For every sequence (x_n) in A such that $x_n \neq c$ for all $n \in \mathbb{N}$ and $\lim_{n \rightarrow \infty} x_n = c$, the sequence of function values $(f(x_n))$ converges to L :

Let $x_n := \frac{1}{n^2}$. By squeeze $-\frac{1}{n} < x_n < \frac{1}{n} \implies \lim x_n = 0$.

Observe the following:

$$\lim_{x \rightarrow 0^+} \frac{1}{\sqrt{x}} = L = \lim_{n \rightarrow \infty} \frac{1}{\sqrt{\frac{1}{n^2}}} = \infty$$

This is not possible as the right hand limit does not evaluate, thus fails the sequential criterion. Therefore is not a convergent limit. ■

$$(d) \quad \lim_{x \rightarrow 0} \sin\left(\frac{1}{x^2}\right)$$

$$f(x) := \sin\left(\frac{1}{x^2}\right)$$

Let the following subsequences be defined as:

$$x_n := \frac{1}{\sqrt{n\pi}} \qquad y_n := \frac{1}{\sqrt{\frac{\pi}{2} + 2\pi n}}$$

Note their convergence:

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n\pi}} = 0 \quad \lim_{n \rightarrow \infty} \frac{1}{\sqrt{\frac{\pi}{2} + 2\pi n}} = 0$$

By sequential criterion $\lim_{n \rightarrow \infty} f(x_n) = \lim_{n \rightarrow \infty} f(y_n)$

Observe $f(x_n), f(y_n)$

$$f(x_n) = \sin \left(\frac{1}{\left(\frac{1}{\sqrt{n\pi}} \right)^2} \right) \quad f(y_n) = \sin \left(\frac{1}{\left(\frac{1}{\sqrt{\frac{\pi}{2} + 2\pi n}} \right)^2} \right)$$

Observe:

$$\begin{aligned} \sin \left(\frac{1}{\left(\frac{1}{\sqrt{\frac{\pi}{2} + 2\pi n}} \right)^2} \right) &= \sin \left(\frac{1}{\left(\frac{1}{\frac{\pi}{2} + 2\pi n} \right)} \right) = \sin \left(\frac{1}{\left(\frac{1}{\frac{\pi}{2}} \right)} \right) = \sin \left(\frac{\pi}{2} \right) \implies \lim_{n \rightarrow \infty} f(y_n) = 1 \\ \sin \left(\frac{1}{\left(\frac{1}{\sqrt{n\pi}} \right)^2} \right) &= \sin \left(\frac{1}{\left(\frac{1}{n\pi} \right)} \right) = \sin(n\pi) \implies \lim_{n \rightarrow \infty} f(x_n) = 0 \end{aligned}$$

$$\lim f(y_n) \neq \lim f(x_n) \implies \lim f(x) \text{ DNE}$$

■

Section 4.1: 14 Let $c \in \mathbb{R}$ and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be such that

$$\lim_{x \rightarrow c} (f(x))^2 = L.$$

(a) Show that if $L = 0$, then

$$\lim_{x \rightarrow c} f(x) = 0.$$

(b) Show by example that if $L \neq 0$, then f may not have a limit at c .

Let $\epsilon > 0$ be given. Since $\epsilon > 0$, it follows that $\epsilon^2 > 0$.

Because we are given that $\lim_{x \rightarrow c} (f(x))^2 = 0$, we know by definition that for any positive number (including ϵ^2), there exists a $\delta > 0$ such that if $0 < |x - c| < \delta$, then:

$$|(f(x))^2 - 0| < \epsilon^2$$

This simplifies to:

$$|f(x)|^2 < \epsilon^2$$

Because both sides are non-negative, we can take the square root to get:

$$|f(x)| < \epsilon$$

Since we have shown that for any $\epsilon > 0$, there exists a $\delta > 0$ such that $0 < |x - c| < \delta \implies |f(x) - 0| < \epsilon$, we can conclude that $\lim_{x \rightarrow c} f(x) = 0$.

Consider the function

$$g(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -1 & \text{if } x < 0 \end{cases}$$

Let $f(x) := g(x)$. Notice that $\lim_{x \rightarrow 0} (f(x))^2 = 1$, since $f(x)$ approaches either 1 or -1 as x approaches 0 from the right or left, and $1^2 = (-1)^2 = 1$. However, $\lim_{x \rightarrow 0} f(x)$ does not exist, as $x \rightarrow 0^+$, $f(x) \rightarrow 1$, but as $x \rightarrow 0^-$, $f(x) \rightarrow -1$.

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